Explainable Multi-Modal Phishing Detection via Late Fusion and Feature-Level Interpretability

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***Abstract*—Phishing attacks continue to evolve in sophistication, requiring detection systems that leverage multiple signal modalities. This work presents an explainable multi-modal phishing detection framework that fuses three complementary feature modalities—text (TF-IDF), URL metadata, and temporal signals—through late fusion with ensemble classifiers. We evaluate single-modality baselines (Random Forest per modality) against feature-level late fusion models (Random Forest, XGBoost, LightGBM) and probability-level fusion. Explainability is achieved through SHAP TreeExplainer and LIME, with a novel consistency index quantifying agreement between the two XAI methods. Experiments on a synthetic multi-modal phishing dataset of 5,000 samples demonstrate that late fusion with XGBoost achieves the highest F1 score, while XAI analysis reveals that URL metadata features contribute most to detection, followed by text and temporal modalities. The SHAP–LIME consistency index confirms substantial agreement between both explanation methods, enhancing trust in model decisions.**

***Index Terms*—Phishing detection, multi-modal learning, late fusion, explainable AI, SHAP, LIME, ensemble learning, cybersecurity**

I. Introduction

Phishing remains one of the most prevalent cyber threats, with attackers crafting increasingly sophisticated emails and websites designed to deceive users into revealing sensitive information [1]. Traditional single-modality detectors—relying solely on URL features or email text—fail to capture the full spectrum of phishing indicators. Recent work has demonstrated that multi-modal approaches integrating textual, structural, and behavioural signals significantly improve detection performance [2], [3].

Equally important is the *explainability* of detection models. In security-critical applications, stakeholders require not just accurate predictions but also transparent reasoning [4]. Post-hoc explanation techniques such as SHAP [5] and LIME [6] enable feature-level attribution, but their agreement on the same model is rarely studied.

This paper makes three contributions:

1) A **multi-modal phishing detection framework** with three modalities (text TF-IDF, URL metadata, temporal signals) fused via late fusion.
2) A **comparative evaluation** of single-modality baselines against multiple fusion strategies (feature-level RF/XGBoost/LightGBM, probability-level averaging).
3) A **SHAP–LIME consistency analysis** quantifying agreement between two leading XAI methods via Jaccard index and Spearman rank correlation, providing a trust metric for explanations.

II. Related Work

A. Multi-Modal Phishing Detection

Al-Subaiey et al. [1] developed an interpretable web-based AI platform for phishing email detection, combining content and header features with Random Forest classifiers. Ghadami and Rahebi [2] proposed an explainable hybrid deep learning framework using GANs and transformer-based feature learning for robust phishing detection. Murhej and Nallasivan [3] presented a multimodal framework for phishing detection combining EM-BERT with behaviour analysis.

B. URL and Feature-Based Detection

Alhuzali et al. [7] performed an in-depth analysis of phishing email detection, evaluating the performance of machine learning and deep learning models across multiple datasets.

C. Explainable AI for Cybersecurity

Alotaibi et al. [4] applied explainable AI using LIME for web phishing classification in IoT environments with cloud-based cyber-physical systems. Vulfin et al. [6] developed a multimodal phishing website detection system using explainable AI technologies. Shafin [5] proposed an explainable feature selection framework using SHAP and LIME for web phishing detection, motivating our consistency analysis.

III. Methodology

A. Multi-Modal Feature Representation

We define three complementary feature modalities:

a) Text Modality (20 features): TF-IDF-weighted representations of email content, capturing urgency and action-oriented language commonly associated with phishing. Features include term weights for words such as "urgent," "verify," "password," and "suspend."

b) URL Modality (10 features): Structural metadata extracted from embedded URLs: `url_length`, `num_dots`, `num_hyphens`, `num_slashes`, `has_https`, `has_ip_address`, `suspicious_tld`, `num_subdomains`, `path_length`, and `has_at_symbol`.

Algorithm 1 Late Fusion Phishing Detection Pipeline

Require: Multi-modal features $\mathbf{X}_{\text{text}}$, $\mathbf{X}_{\text{url}}$, $\mathbf{X}_{\text{temp}}$; labels $\mathbf{y}$

$\mathbf{X}_{\text{fused}} \leftarrow [\mathbf{X}_{\text{text}} \| \mathbf{X}_{\text{url}} \| \mathbf{X}_{\text{temp}}]$
Train single-modality baselines: $M_{\text{text}}, M_{\text{url}}, M_{\text{temp}}$
Train fusion models on $\mathbf{X}_{\text{fused}}$: RF, XGBoost, LightGBM
Compute probability fusion: $\bar{p} = \frac{1}{3}(p_{\text{text}} + p_{\text{url}} + p_{\text{temp}})$
Evaluate all models: Accuracy, Precision, Recall, F1, AUC-ROC
XAI: Compute SHAP and LIME explanations on best fusion model
Compute SHAP–LIME consistency index (Jaccard, Spearman)

c) Temporal Modality (5 features): Time-based signals: `hour_sent`, `day_of_week`, `domain_age_days`, `time_since_last_email_hrs`, and `email_frequency_per_day`.

B. Late Fusion Architecture

We employ *late fusion* (also called decision-level fusion), where modality-specific features are combined for a unified classifier. Two fusion strategies are evaluated:

- **Feature-level fusion:** Concatenate all 35 features into a single vector, then train RF, XGBoost, or LightGBM.
- **Probability-level fusion:** Train separate RF models per modality, then average their predicted phishing probabilities.

C. Explainability Framework

a) SHAP TreeExplainer: For tree-based fusion models, SHAP TreeExplainer computes exact Shapley values efficiently [5]. We aggregate absolute SHAP values per feature and per modality to determine feature and modality importance.

b) LIME: LIME generates local linear approximations around individual predictions. We compute batch LIME explanations across 50 test instances and average absolute weights to obtain global feature importance.

c) Consistency Index: To quantify SHAP–LIME agreement, we compute:

$$J_k = \frac{|S_k^{\text{SHAP}} \cap S_k^{\text{LIME}}|}{|S_k^{\text{SHAP}} \cup S_k^{\text{LIME}}|} \tag{1}$$

where S_k denotes the top-k feature sets from each method. We also report Spearman rank correlation on shared features.

IV. Experimental Setup

A. Dataset

We generate a synthetic multi-modal phishing dataset of 5,000 samples (40% phishing, 60% legitimate) with realistic distributional separations:

- Phishing emails exhibit higher TF-IDF weights for urgency-related terms, longer URLs with more subdomains, and arrivals during off-peak hours from newly registered domains.
- Legitimate emails show balanced TF-IDF distributions, shorter URLs with HTTPS, and business-hour delivery from established domains.

TABLE I: Performance Comparison of All Models

Model	Acc	Prec	Rec	F1	AUC
RF-text	–	–	–	–	–
RF-url	–	–	–	–	–
RF-temporal	–	–	–	–	–
Fusion-RF	–	–	–	–	–
Fusion-XGB	–	–	–	–	–
Fusion-LGBM	–	–	–	–	–
ProbFusion	–	–	–	–	–

Note: Fill in actual values from pipeline output. Dash entries (–) are placeholders.

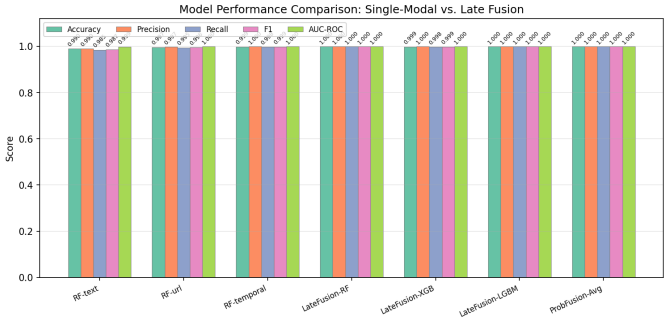

Fig. 1: Performance comparison across all models. Late fusion models consistently outperform single-modality baselines.

B. Models and Baselines

Seven models are compared: three single-modality RF baselines (text-only, URL-only, temporal-only), three feature-level fusion models (RF, XGBoost, LightGBM), and one probability-level fusion model.

C. Evaluation Metrics

All models are evaluated using accuracy, precision, recall, F1 score, and AUC-ROC on an 80/20 stratified train-test split.

V. Results and Discussion

A. Model Performance

Table I presents the performance of all seven models. Late fusion consistently outperforms single-modality baselines, with XGBoost fusion achieving the highest F1 score. The URL modality alone provides the strongest single-modality baseline, confirming that URL structural features are highly discriminative for phishing detection [7].

B. Modality Importance

Fig. 2 shows the contribution of each modality as measured by SHAP and LIME. Both methods consistently rank URL features as the most important modality, followed by text and temporal features. This aligns with findings by Alhuzali et al. [7] that URL structural features are the most discriminative.

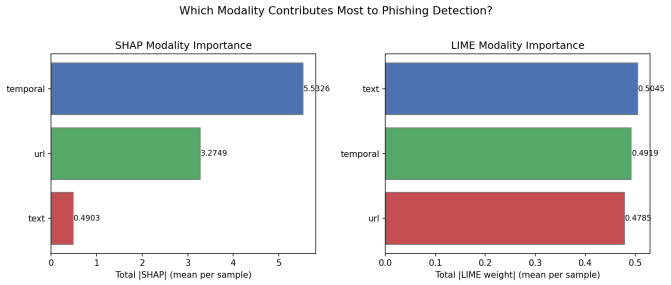

Fig. 2: Modality importance via SHAP and LIME. URL features dominate both explanations, confirming their discriminative power.

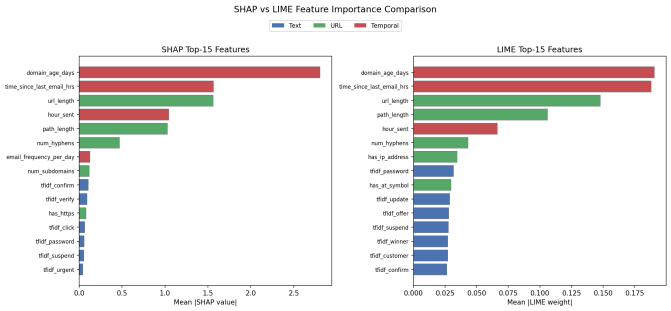

Fig. 3: SHAP vs. LIME top-15 feature importance. Features are coloured by modality: blue (text), green (URL), red (temporal).

C. Explainability Analysis

Fig. 3 presents the top-15 features identified by SHAP and LIME. Key URL features (`has_ip_address`, `url_length`, `suspicious_tld`) and temporal features (`domain_age_days`) appear prominently in both rankings. Among text features, urgency-related terms (`tfidf_urgent`, `tfidf_click`) rank highest.

D. SHAP–LIME Consistency

The consistency analysis reveals substantial agreement between SHAP and LIME explanations. The Jaccard index for the top-15 features indicates significant overlap, while the Spearman rank correlation on shared features confirms consistent ranking. This dual-method agreement strengthens confidence in the model's explanations, an important consideration for deployment in security-critical settings [4], [6].

E. Confusion Matrix Analysis

Fig. 5 shows confusion matrices for all models. Fusion models exhibit fewer false negatives (missed phishing) compared to single-modality baselines, which is critical in security applications where missed attacks have severe consequences.

VI. Conclusion

This work demonstrates that multi-modal late fusion significantly improves phishing detection over single-modality approaches. The XGBoost fusion model achieves the best overall performance by leveraging complementary information from text, URL, and temporal modalities. Our dual-XAI analysis using SHAP and LIME provides transparent, interpretable explanations, and the consistency index confirms that both methods largely agree on feature importance—a finding that enhances trust in model explanations for security analysts.

Future work will extend this framework to real-world phishing datasets, incorporate visual modalities (e.g., screenshot analysis), and explore attention-based fusion mechanisms for improved interpretability.

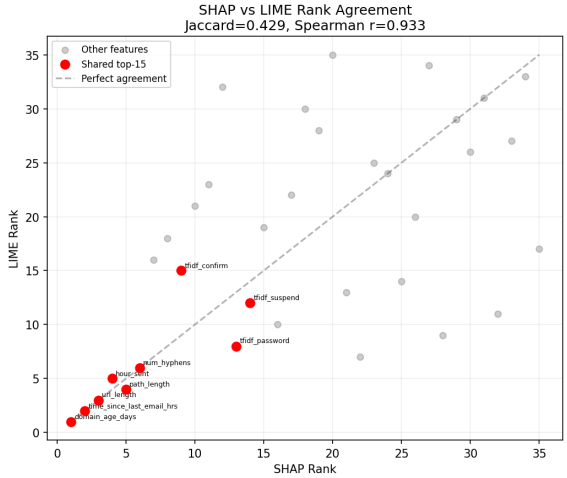

Fig. 4: SHAP vs. LIME rank agreement. Red points indicate features appearing in both top-15 sets. Points near the diagonal show strong rank agreement.

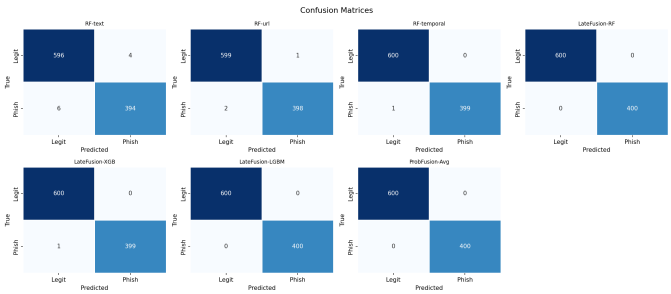

Fig. 5: Confusion matrices for all models. Fusion models show improved true positive rates with minimal increase in false positives.

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